e^3

Learning to Explore Enables Extrapolation of Test-Time Compute for LLMs

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Intro

Test-time scaling

최근 reasoning LLM에서는 inference 때 더 많은 compute를 쓰게 하면 (= 더 긴 reasoning trace를 생성하게 하면) 성능이 좋아지는 **test-time scaling**이 중요한 방향으로 여겨짐

Extrapolation

저자들은 test-time scaling에서 중요한 것은 단순히 “긴 답변을 생성하면 성능이 좋다”가 아니라고 봄!

Extrapolation: 학습할 때 사용한 최대 reasoning token budget보다 더 많은 토큰을 inference에서 줬을 때도 성능이 계속 좋아지는가?

Test-time compute extrapolation

모델이 단순히 훈련 시 본 길이 안에서 최적화된 것인지, 아니면 실제로 “시간을 더 주면 더 많은 탐색을 해서 답을 찾아내는 방법”을 배운 것인지 확인하려 함

Research problem

Training objective: Optimization

Outcome-reward RL or SFT optimizes reasoning traces under a fixed maximum output token length (training budget).

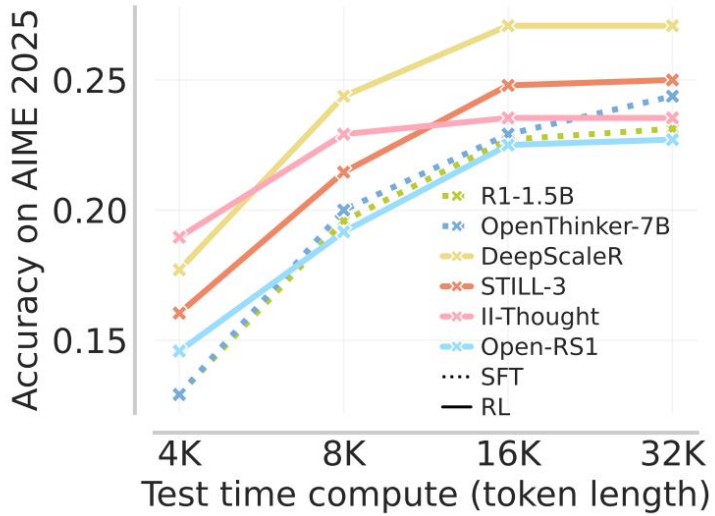
Deployment objective: Extrapolation

Accuracy should keep improving when the test budget exceeds training budget, while remaining inside the pretrained context window.

Problem

Most reasoning models gain little accuracy when inference length exceeds the post-training budget.

More compute ⇒ longer output ⇏ higher accuracy



Method Overview

Hypothesis

Extrapolation requires a model to explore inside its own response by chaining generation, verification, and revision.

In-context exploration

Recipe

Base-model asymmetry

Verification can be more reliable than initial answer generation.

Negative gradients

Failed traces lose probability, including early EOS decisions.

Coupled curriculum

Problem difficulty and output budget increase together.

Extrapolation

Longer test budgets support additional search and higher accuracy.

e^3 is a post-training recipe, not a new Transformer architecture or a new inference-time search algorithm.

KEY CONCEPT

In-context exploration

Unproductive length

- Restating the same derivation
- Repeating tokens after the original answer
- Adding generic reflection without changing the hypothesis

Productive exploration

- Generate a candidate solution
- Check a calculation or assumption
- Condition the next attempt on earlier failures



METHOD

Ingredient 1: asymmetric capabilities

Verification-generation gap

The model can judge a candidate more reliably than it can generate a correct candidate on the first attempt.

Why it matters

- A failed attempt still supplies information
- Verification can guide the next attempt
- Additional tokens can increase the number of useful attempts

Simplified model

Each attempt fails with probability p

Failure after k independent attempts

$$p^k$$

Success = $1 - p^k$

EXPERIMENTAL DESIGN

Controlled tests of the VG gap

Countdown

- Search for an arithmetic expression matching a target
- **Generation is difficult, but equation verification is easy**
- 40,000 examples across 3 to 6 candidate numbers

Mult and Mult-V

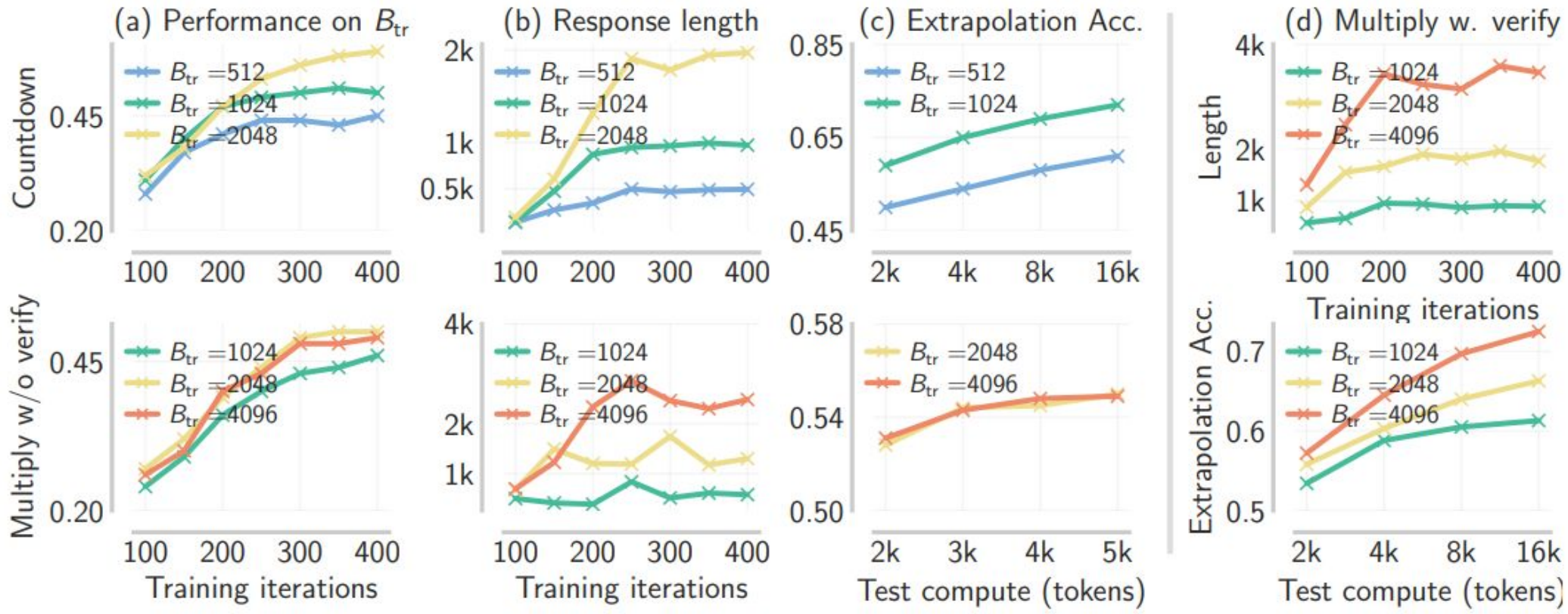
- Five-digit multiplication has a **weak natural VG gap**
- Mult-V first receives verification-rich SFT traces
- RL then tests whether verification enables longer search

Expected contrast

If asymmetry matters, Countdown and Mult-V should gain more from longer budgets than Mult.

RESULTS

Asymmetry enables length utilization



Countdown

Longer budgets increase response length and extrapolation accuracy.

Mult

Length and extrapolation plateau when verification is weak.

Mult-V

Verification-rich initialization restores upward scaling trends.

Ingredient 2: negative policy gradients

What is *Negative gradient*?

$$\mathbb{E}_{\boldsymbol{y}\sim\tilde{\pi}(\cdot|\boldsymbol{x})}\left[A_i(\boldsymbol{x},\boldsymbol{y})\cdot\nabla_\pi\log\pi(\boldsymbol{y}\mid\boldsymbol{x})\right]$$ (simple form of policy gradient in RL)

- x: prompt/problem
- y: 모델이 생성한 reasoning trace
- π: 현재 학습하는 policy
- $\tilde{\pi}$: rollout을 실제로 생성한 sampling policy
- $A(x,y)$: advantage
- $\log\pi(y|x)$: 그 response에 현재 model이 부여하는 log-probability

Advantage function: 특정 상태(s)에서 어떤 행동(a)이 평균적으로 기대되는 행동보다 얼마나 더 좋은지 그 우수성을 수치로 나타낸 값

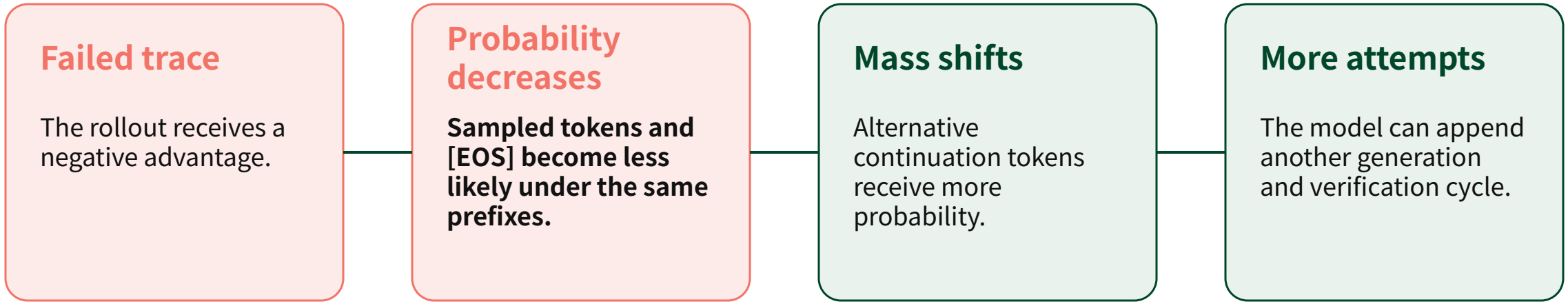
RL에서는 response가 얼마나 좋은지에 따라 advantage가 달라짐

Correct response: A(x, y) > 0 ➡ ∇ log π(y | x) 방향으로 업데이트 (= 이 response를 더 자주 생성해라)

Incorrect response: **A(x, y) < 0 ➡ Negative gradient** (= 이 response의 probability를 낮춰라)

METHOD

Ingredient 2: negative policy gradients

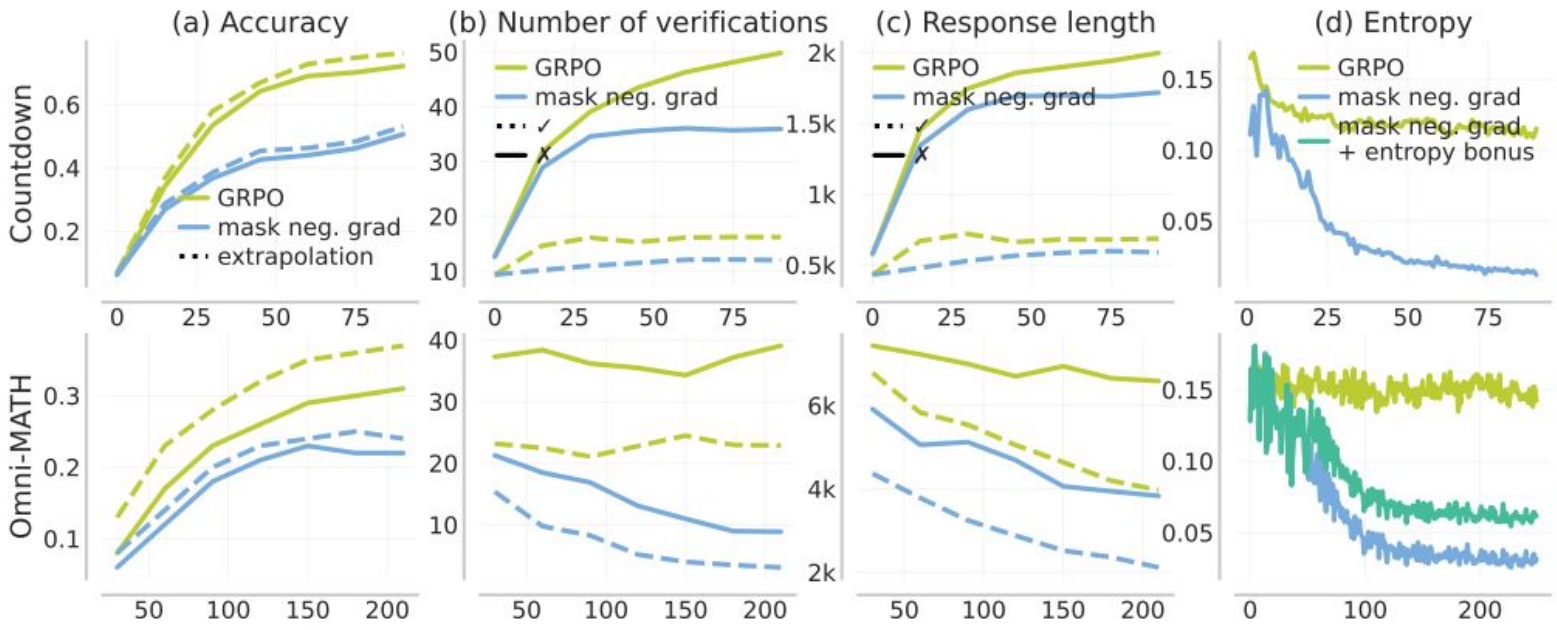


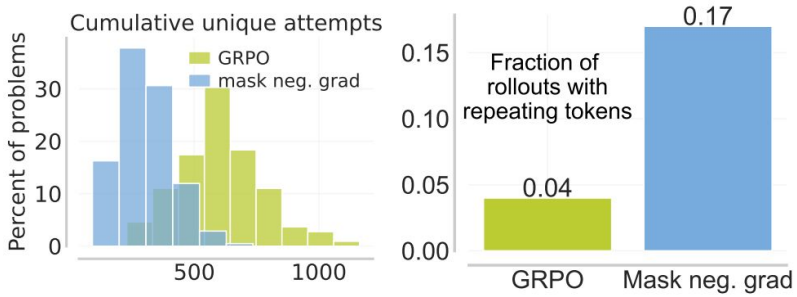
GRPOMask ablation

Mask negative-advantage updates while keeping positive updates.
Differences isolate the contribution of failed traces more directly than an RL-versus-SFT comparison.

RESULTS

Negative gradients increase diversity and chaining





4%
repetition with GRPO

17%
when negative gradients are masked

- GRPO maintains higher entropy and longer traces
- Verification attempts rise when the base model has a VG gap
- The performance gap widens when the test budget doubles on hard DMath

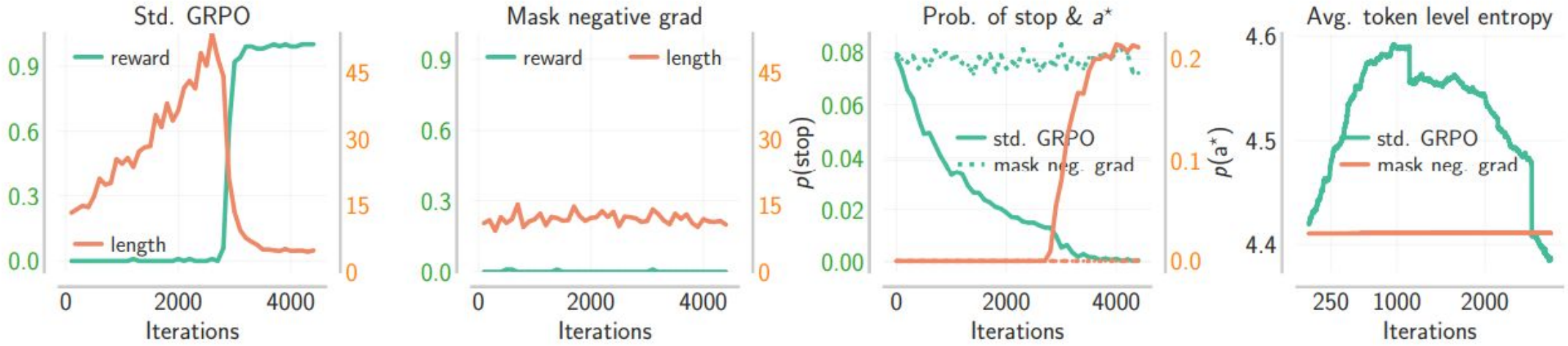
The p^k model separates exploration from sharpening

Toy MDP

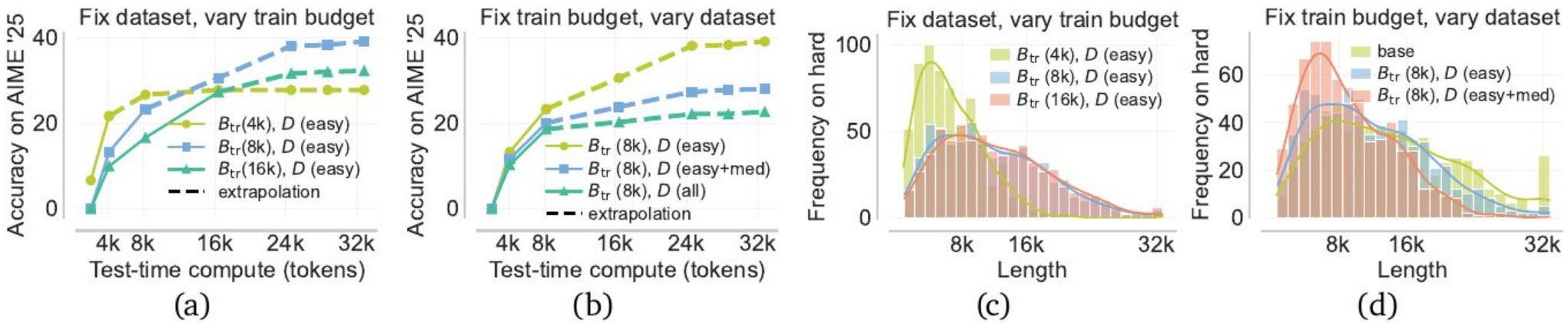
- 100 actions plus a stop action
- The correct action a* initially has very low probability
- Perfect verification ends the rollout immediately after a*

Observed dynamics

- Negative gradients lower p(stop)
- Average length grows from about 15 to 45
- After discovering a*, positive gradients sharpen the policy



Fixed budgets and static data mixtures are insufficient



Budget too short

Long successful traces exceed the limit and receive zero reward.

Budget too long

Long-horizon RL becomes unstable and high-variance.

Hard data too early

The model commits early because exploration cannot finish inside the budget.

METHOD

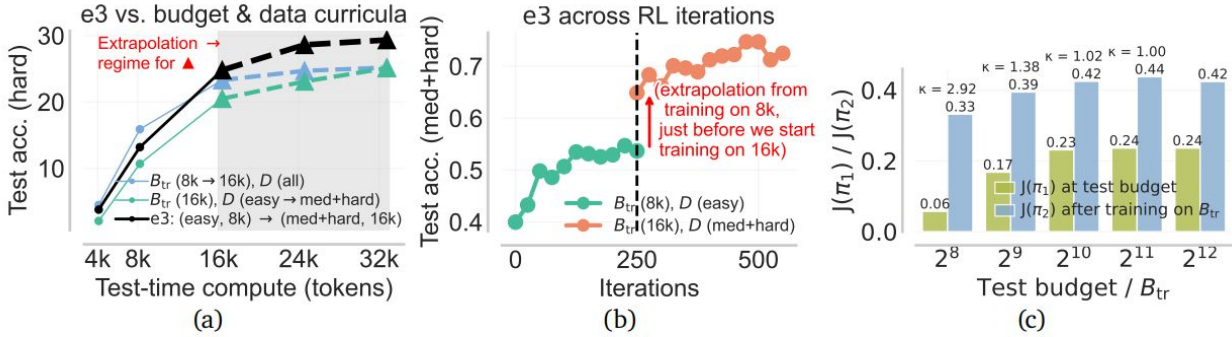
Ingredient 3: coupled data and budget curriculum

Budget selection rule

$$B^{\star}_{\mathrm{tr},i}(D_i) = \arg\min_{B \geq B_0} B$$

subject to $J(\pi_i; D_i, 2 \cdot B) \leq \kappa \cdot J(\pi_i; D_i, B), \quad \kappa > 1$

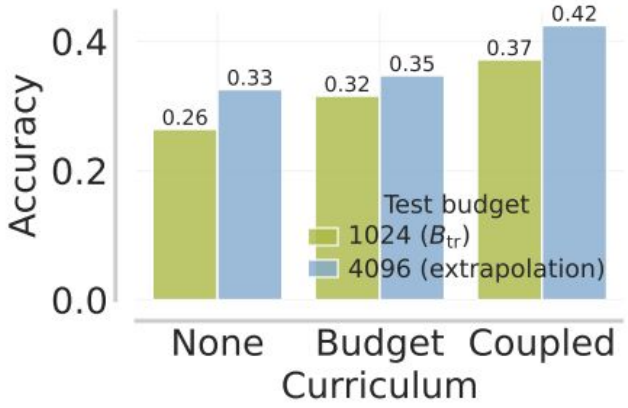
κ = 1.2 | candidates: 4k, 8k, 16k



Interpretation

Choose the smallest practical budget after which doubling the length yields at most a 20% relative gain.

Countdown accuracy



Final e^3 training schedule

Stage 1

- Qwen3-1.7B
- Easy DMath problems
- 8k output-token budget

Stage 2

- Continue from Stage 1
- Medium + hard DMath
- 16k output-token budget

Deployment evaluation

Test budgets up to 32k, twice the maximum RL training budget

EVALUATION

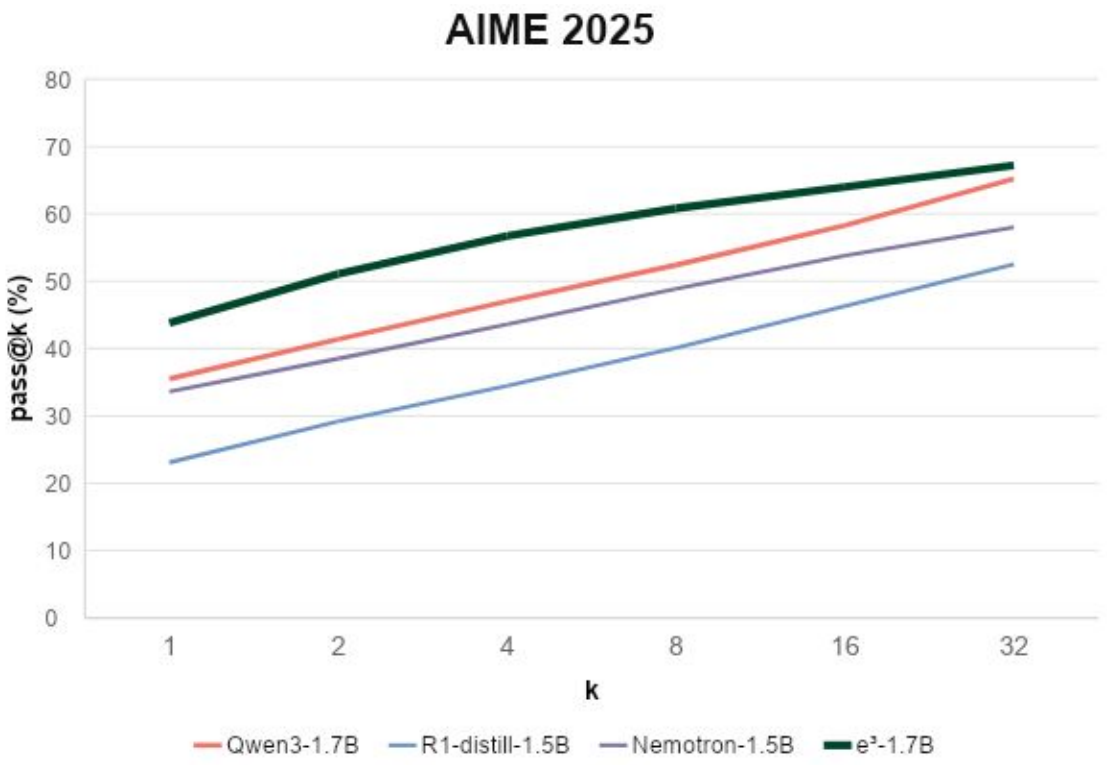
Performance evaluation

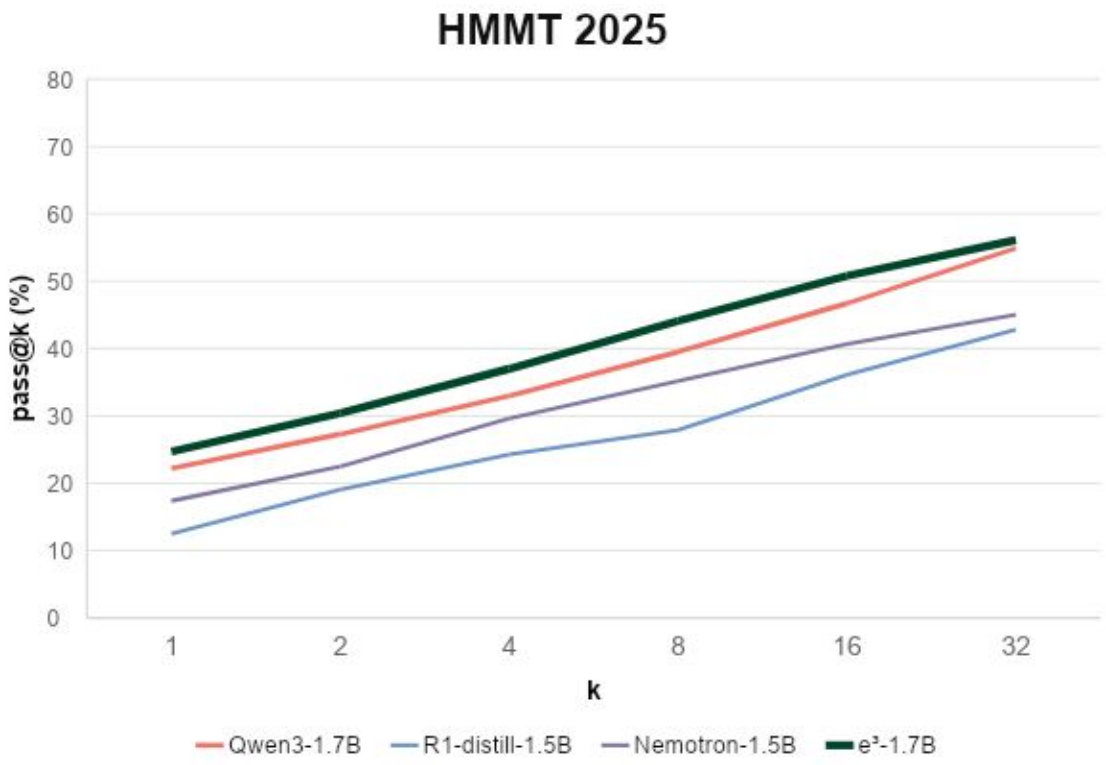
Question	Protocol
Length extrapolation	Evaluate at 4k, 8k, 16k, 24k, and 32k
OOD reasoning	AIME 2025 and HMMT 2025
Sampling	128 rollouts per prompt for final pass@k estimates
Metrics	pass@1 and pass@k for k = 2, 4, 8, 16, 32
Baselines	Qwen3-1.7B, R1-distill-Qwen-1.5B, Nemotron-Reasoning-1.5B
Length forcing	Append ‘Wait’ multiple times to competing models

The paper uses output-token count as a proxy for compute. It does not normalize FLOPs or latency across model sizes.

FINAL RESULTS

e^3 improves pass@k on both 2025 benchmarks





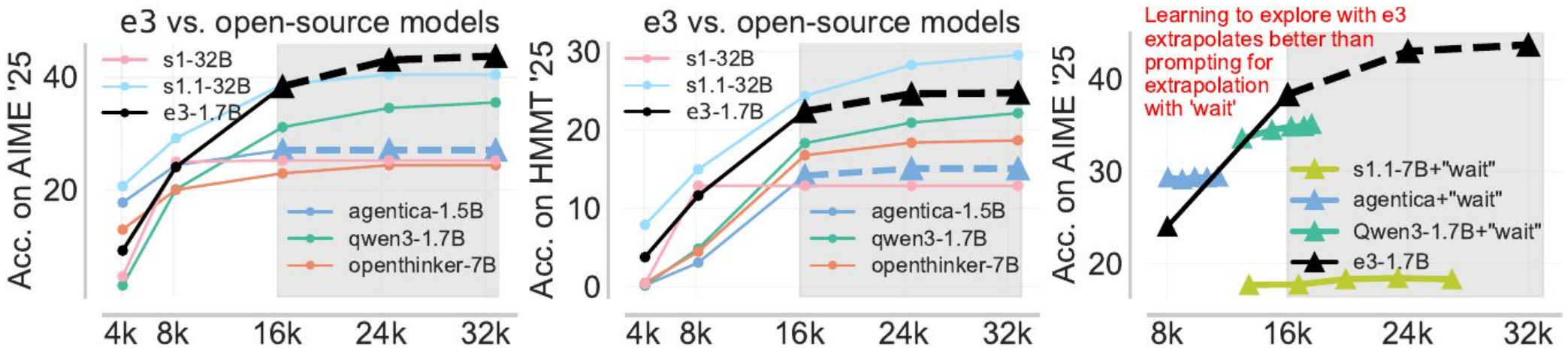
AIME pass@1: +8.3 points

HMMT pass@1: +2.5 points

Base comparison: Qwen3-1.7B

FINAL RESULTS

e^3 scales better than simple budget forcing



- The 16k-trained e^3 model continues improving at 24k and 32k
- On AIME 2025, e^3-1.7B exceeds OpenThinker-7B and s1.1-32B at long budgets
- Appending ‘Wait’ extends output but produces weaker scaling

Strengths and stated limitations

Strengths

- A clear definition of extrapolation beyond maximum training budget
- Controlled tasks, ablations, and a toy theory support one mechanism
- The final model improves both pass@1 and pass@32
- The curriculum rule is practical and easy to test

Author-stated limitations

- Final evaluation is limited to a 1.7B model and mathematics
- Most experiments study only the verification-generation gap
- Repetition bias blocks further length scaling
- The need for curricula at larger model scales remains unclear

Novelty lies in the mechanism and recipe, not in a new architecture or RL objective.

INTERPRETATION

What the evidence supports

Claim	Evidence strength
Negative gradients increase length and diversity	Relatively strong
A VG gap helps RL exploit longer budgets	Moderate
Coupled curricula outperform static settings	Moderate to strong
e^3 improves the evaluated 1.7B model	Strong within reported benchmarks
Asymmetry is universally required	**Not established**

Review **The empirical recipe is convincing. The paper's universal mechanistic language is stronger than the evidence.**

Concerns: causal mechanism

1 Mult-V changes more than verification

Verification-rich SFT can also change arithmetic skill, response length, style, and entropy. The experiment does not isolate the VG gap alone.

2 Verification counts are surface-level proxies

Blank lines, ‘Check:’, and ‘Wait’ do not prove that the model detected and corrected an error. Longer traces mechanically contain more segments.

3 GRPOMask changes more than one effective optimization property

Masking negative advantages changes the overall optimization dynamics, so the ablation does not perfectly isolate the semantic role of learning from failed traces.

RESEARCH EXTENSION

Method improvements and new research directions

Matched asymmetry intervention

Create paired traces with identical calculations, length, and style. Add or remove only semantically valid verification steps.

Adaptive token allocation

Learn a budget controller that expands the response only when verification predicts that another attempt has positive value.

Process-aware reward

Reward error detection, correction quality, and non-repetition instead of relying only on final-answer reward.

Beyond verification-generation

Test retrieval-generation, planning-execution, critique-revision, and simulation-decision asymmetries across domains.

Research question

Which asymmetries create genuinely new solution support rather than longer reformulations?

“단순히 길게 생각하게만 만드는 비대칭”과 “진짜로 새로운 풀이를 찾게 만드는 비대칭”을 구분하자

TAKEAWAYS

Conclusion

1 **Useful test-time scaling requires learned exploration behavior, not only longer outputs.**

2 **e^3 combines base-model asymmetry, negative gradients, and a coupled curriculum to train that behavior.**

3 **The 1.7B results are strong, but the universal mechanism still needs cleaner causal tests, uncertainty estimates, and broader domains.**

Empirically promising recipe. Mechanistic claim remains provisional.